```c
#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"
#include "xuartlite.h"
#include "xparameters.h"


XUartLite_Config *uart_config1,*uart_config2;
XUartLite uart1,uart2;


void uart_init(){

        uart_config1 = XUartLite_LookupConfig(XPAR_UART0_DEVICE_ID);
        int status1 = XUartLite_CfgInitialize(&uart1,uart_config1, uart_config1->RegBaseAddr);

        uart_config2 = XUartLite_LookupConfig(XPAR_UART1_DEVICE_ID);
    int status2 = XUartLite_CfgInitialize(&uart2,uart_config2, uart_config2->RegBaseAddr);

        if((status1 && status2) == XST_SUCCESS)
                xil_printf("UART INIT SUCCESSFUL\n");
        else
                xil_printf("UART INIT FAILED\n");


}

int main()
{
    init_platform();
     uart_init();

        u8 data1[] ="UART 0->1";
        u8 data2[] ="UART 1->0";
        u8 Rx[9];

        xil_printf("Sending Data from UART 0 -> UART 1 \n");

        XUartLite_Send(&uart1, &data1[0], 9);
        while(XUartLite_IsSending(&uart1));


        int byteRcvd = 0;

        while(byteRcvd != 9) {
           byteRcvd = byteRcvd + XUartLite_Recv(&uart2, &Rx[byteRcvd], 9);
           }


         xil_printf("Transmission Complete UART0 -> UART1\n");

           for(int i = 0 ; i < 9; i++)
           {
                xil_printf("%0c",Rx[i]);
           }
           xil_printf("\n");
           /////////////////////////////////////////////////

           xil_printf("Sending Data from UART 1 -> UART0\n");


           XUartLite_Send(&uart2, &data2[0], 9);
           while(XUartLite_IsSending(&uart2));

           byteRcvd = 0;

           while(byteRcvd != 9) {
              byteRcvd = byteRcvd + XUartLite_Recv(&uart1, &Rx[byteRcvd], 9);
              }

             xil_printf("Transmission Complete UART1 -> UART0\n");

              for(int i = 0 ; i < 9; i++)
              {
                xil_printf("%0c",Rx[i]);
              }
              xil_printf("\n");




    cleanup_platform();
    return 0;
}
```